Natanael S. Alpay

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Education

University of California, Irvine 2022–Present

Ph.D. in Mathematics, expected May 2027
M.S. in Applied Mathematics, December 2025

Chapman University 2017–2022

B.S. in Mathematics
B.S. in Physics
B.S. in Computer Science
B.A. in French

Certificates and Professional Training

Erdős Institute – Quant Finance Boot Camp Fall 2026

Certificate expected November 13, 2026. Training in option pricing, Monte Carlo simulation, hedging, portfolio risk, and backtesting. Capstone project: *Memory-Efficient Deep Hedging*.

Northeast Big Data Innovation Hub and NSDC Jan 2025

Training in data cleaning, Python, time-series and geospatial analysis, and data visualization.

Academic Appointments and Work Experience

Research Assistant – University of California, Irvine Sep 2022–Present

Investigate problems in probability, harmonic and complex analysis, optimization, and machine learning, combining mathematical proofs with reproducible numerical experiments.

Teaching Assistant – University of California, Irvine Sep 2022–Present

Support undergraduate courses in calculus, linear algebra, real and complex analysis, differential equations, geometry, optimization, numerical analysis, and mathematical machine learning. Current assignment: MATH 140A, Real Analysis (Fall 2026).

Instructor of Record – University of California, Irvine Summer 2026

Taught MATH 105A, Numerical Analysis I (Summer Session I), and MATH 105B, Numerical Analysis II (Summer Session II), with corresponding numerical analysis laboratories.

Directed Reading Program Mentor – UC Irvine Fall 2026–Present

Lead undergraduate reading and research on Deep Hedging and Deep Bellman Hedging, connecting probability and optimization with Python experiments and critical evaluation of neural hedging strategies.

Undergraduate Research Lead – UC Irvine Apr 2025–Present

Guided eight students across two projects on high-dimensional probability and random matrices, including numerical investigations of norm preservation and the preparation of two research posters.

Undergraduate Research Mentor – DATA@UCI Jan–Jun 2025

Mentored an undergraduate team in Python data analysis, dimensionality reduction, research design, and presentation.

Research Assistant – Chapman University Jun–Aug 2022

Implemented and evaluated machine-learning algorithms, connecting their mathematical foundations with computational experiments.

Supplemental Instructor – Chapman University Sep 2020–May 2022

Facilitated supplemental instruction for Calculus II (MATH 111).

Tutor – Chapman University Sep 2019–May 2022

Tutored students in mathematics, physics, French, and computer science.

Student Undergraduate Research Assistant 2021

Conducted research in the Department of Mathematics at Chapman University.

Student Assistant – Chapman University 2018–Dec 2021

Assisted with calculus instruction and grading.

Publications and Preprints

- N. Alpay. *Sharp discrete Hardy constants in low dimensions*. Preprint, arXiv:2610.05421 (2026).
- N. Alpay. *The sharp discrete Hardy inequality on* $\mathbb{Z}^3$. Preprint, arXiv:2608.25262 (2026).
- N. Alpay, P. Cerejeiras, and U. Kähler. *Generalized Fock-space and fractional derivatives: Uniqueness of sampling and interpolation sets*. *Results in Applied Mathematics* **31** (2026), article 100737.
- N. Alpay. *Sharp Ternary Martingale Isoperimetry and n-adic Takagi-Type Lower Bounds*. Preprint, arXiv:2607.11069 (2026).
- N. Alpay and E. Battaglia. *Complex Stochastic Gradient Descent and Directional Bias in Reproducing Kernel Hilbert Spaces*. Preprint, arXiv:2604.23017 (2026).
- N. Alpay, A. De Martino, and K. Diki. *Representer Theorem in Complex Reproducing Kernel Hilbert Spaces with Applications to Fock and Hardy Spaces and Superoscillations*. Preprint, arXiv:2604.22165 (2026).
- N. Alpay. *Fractional Supershifts and their associated Cauchy Evolution problems*. Accepted for publication in *Proceedings of the American Mathematical Society* (June 2026). Preprint: arXiv:2601.11829.
- N. Alpay, A. De Martino, and K. Diki. *The Fueter Mittag-Leffler Bargmann transform*. *Journal of Mathematical Analysis and Applications* **549** (2025), no. 2, article 129480.
- N. Alpay and T. C. John. *Thermal states on Mittag-Leffler Fock space of the slitted plane*. *Journal of Mathematical Analysis and Applications* **547** (2025), no. 1, article 129314.
- N. Alpay and P. Ivanisvili. *Sharp Lower Bounds for Dyadic Square Functions of Indicator Functions of Sets*. Preprint, arXiv:2502.16045 (2025; revised 2026).
- N. Alpay, P. Jipsen, and M. Sugimoto. *Varieties of unary-determined distributive ℓ-magmas and bunched implication algebras*. *Logical Methods in Computer Science* **20** (2024), no. 1, article 12.

- N. Alpay and K. Diki. *On the Mittag Leffler Bargmann (MLB) transform. Journal of Mathematical Analysis and Applications* **527** (2023), no. 2, article 127458.
- N. Alpay. *A new characterization of the Hardy space and of other Hilbert spaces of analytic functions. Istanbul Journal of Mathematics* **1** (2023), no. 1, pp. 1–11.
- N. Alpay, P. Jipsen, and M. Sugimoto. *Unary-Determined Distributive ℓ-magmas and Bunched Implication Algebras.* In *Relational and Algebraic Methods in Computer Science (RAMiCS 2021)*, Lecture Notes in Computer Science **13027**, Springer, 2021, pp. 19–36. Conference version of the expanded 2024 journal article.
- N. Alpay and P. Jipsen. *Commutative Doubly-Idempotent Semirings Determined by Chains and by Preorder Forests.* In *Relational and Algebraic Methods in Computer Science (RAMiCS 2020)*, Lecture Notes in Computer Science **12062**, Springer, 2020, pp. 1–14.

Awards and Honors

NSF Travel Support – IWOTA 2026 2026

Travel support for the International Workshop on Operator Theory and its Applications, Université Laval, Québec City, August 3–7, 2026.

NSF Travel Support Grant – IWOTA 2025 Jun 2025

Travel grant for the 36th International Workshop on Operator Theory and its Applications, University of Twente, July 14–18, 2025.

Ronald M. Huntington Outstanding Scholarship Award May 2022

Awarded to the graduating senior judged to have exhibited the most distinguished record of scholarly accomplishment at Chapman University.

Outstanding Mathematics Student Honors May 2022

Recognized as the outstanding mathematics student in the graduating class of Schmid College of Science and Technology, Chapman University.

Outstanding Physics Student Honors May 2022

Recognized as the outstanding physics student in the graduating class of Schmid College of Science and Technology, Chapman University.

Undergraduate Scholarly/Creative Grant Apr 2020

Received support for *The Structure of Distributive Idempotent Weakly Conservative Lattice-Ordered Magmas.*

Summer Undergraduate Research Fellow (SURF) Award Apr 2020

Selected for Chapman University's Summer Undergraduate Research Fellowship for *Differential Forms as Tools in Differential Geometry and Analysis on Manifolds.*

Undergraduate Scholarly/Creative Grant Jun 2019

Received support for *Commutative Doubly-Idempotent Semirings Determined by Chains and by Preorder Forests.*

DataFest – First Place and Best Insight May 2019

Earned team awards at DataFest 2019, Chapman University, as a member of *git rekt.*

Research Talks and Workshops

- **Upcoming, November 2026. UCI Math Finance Club.** Invited workshop on hedging, machine learning, and portfolio optimization.
- August 5, 2026. **International Workshop on Operator Theory and its Applications (IWOTA 2026)**, Université Laval. *On the Complex Representer Theorem and Applications to Minimization Problems Over Reproducing Kernel Hilbert Spaces.*
- August 4, 2026. **IWOTA 2026**, Université Laval. *Lower Bounds for Dyadic Square Functions of Indicator Functions of Sets.*
- August 3, 2026. **IWOTA 2026**, Université Laval. *Generalized Fock-Space and Fractional Derivatives: Uniqueness of Sampling and Interpolation Sets.*
- April 29, 2026. **CHOC Hospital Pediatric and Lifespan Data Science Conference**. *Integrating Machine Learning for VAERS Vaccine Safety Signal Triage vs. Classical Disproportionality: A Pediatric and Lifespan Evaluation.* Podium presentation.
- April 25, 2026. **Southern California Applied Mathematics Symposium (SoCAMS 2026)**. *On the Complex SGD with Application to Kernel Regression in Reproducing Kernel Hilbert Spaces and Directional Bias.*
- January 16, 2026. **Workshop on Applications of Hypercomplex Analysis**. *Fractional Supershifts and Their Associated Cauchy Evolution Problems.*
- January 6, 2026. **Joint Mathematics Meetings**. *Lower Bounds for Dyadic Square Functions of Indicator Functions of Sets.*
- January 4, 2026. **Joint Mathematics Meetings**. *The Mittag-Leffler Bargmann Transform, and Its Extension to the Hypercomplex Setting.*
- July 18, 2025. **IWOTA 2025**, University of Twente. *Thermal States on Mittag-Leffler Fock Space of the Slitted Plane.*
- July 18, 2025. **IWOTA 2025**, University of Twente. *The Mittag-Leffler Bargmann Transform, and Its Extension to the Quaternionic Setting.*
- April 26, 2025. **Southern California Applied Mathematics Symposium**. *Special Functions as Solutions to Least-Squares Problems.*
- March 29, 2025. **22nd Annual Workshop on Applications and Generalizations of Complex Analysis**, University of Aveiro. *Lower Bounds for Dyadic Square Functions of Indicator Functions of Sets.*
- March 20, 2025. **Mathematics of Data, Dynamics, and Life Sciences**. *On the Complex Representer Theorem and Applications to Minimization Problems.*
- November 19, 2024. **Advances in Operator Theory with Applications to Mathematical Physics**. *On the Complex Representer Theorem and Applications.*
- January 11, 2024. **Workshop on Schur Analysis and Applications to Hypercomplex Analysis, Neural Networks and Linear Systems**. *Thermal States on Mittag-Leffler Fock Space of the Slitted Plane.*

- January 3, 2024. **Joint Mathematics Meetings**. *Thermal States on Mittag-Leffler Fock Space of the Slitted Plane.*
- October 31, 2023. **Heat Flow, Two-Point Inequalities, and Beyond**, UC Irvine. *On Complex Hypercontractivity after S. Janson.*
- November 15, 2022. **Advances in Operator Theory with Applications to Mathematical Physics**. *On the Mittag-Leffler Bargmann (MLB) Transform.*
- May 31, 2022. **Workshop on Operator Theory with an Eye on Linear Systems and Hypercomplex Analysis**. *On the Mittag-Leffler Bargmann (MLB) Transform.*
- November 20, 2021. **SCCUR**. *Generalized Fock Space Using Fractional Derivatives with Applications to Uniqueness of Sampling and Interpolation Sets.*
- August 11, 2021. **IWOTA 2021**, special session on Hilbert Spaces of Analytic Functions and Applications. *Generalized Fock Space and Fractional Derivative.*
- October 29, 2020. **18th International Conference on Relational and Algebraic Methods in Computer Science (RAMiCS 2020)**, École polytechnique, Palaiseau. *Commutative Doubly-Idempotent Semirings Determined by Chains and by Preorder Forests.*
- May 9, 2020. **Chapman Capstone Presentation**. *The Structure of Distributive Idempotent Weakly Conservative Lattice-Ordered Magmas.*
- May 9, 2020. **Chapman Capstone Presentation**. *Minimal Surfaces.*
- January 17, 2020. **Joint Mathematics Meetings**, Denver, AMS Contributed Paper Session on Algebra and Algebraic Geometry. *Commutative Doubly-Idempotent Semirings Determined by Chains and by Preorder Forests.*
- November 10, 2019. **AMS Fall Western Sectional Meeting**, UC Riverside, Undergraduate Research Special Session. *Some Structural Results About Commutative Doubly-Idempotent Semirings.*

Poster Presentations

- October 14, 2026. **Symposium on Risks and Opportunities of AI in Pharmaceutical Medicine (AIPM)**, UC Berkeley. *AI-Assisted Prioritization of VAERS Signal Candidates.*
- January 23, 2026. **Biostatistics Symposium of Southern California**. *Integrating Machine Learning for Pharmacovigilance Adverse Event Signal Detection Compared to Classical Disproportionality.*
- August 4, 2021. **Schmid Summer Research Showcase**, Chapman University. *Machine Learning and Neural Networks: The Theory Behind Implementations.*
- May 6, 2021. **Student Symposium**, Chapman University. *Quantum Trajectories for Qutrit Processes.*
- May 6, 2021. **Student Symposium**, Chapman University. *The Mathematical Basis of General Relativity.*
- November 27, 2019. **Student Scholar Symposium**, Chapman University. *Some Structural Results About Commutative Doubly-Idempotent Semirings.*

Current Research Projects

Memory-Efficient Deep Hedging 2026–Present

Develop Python/PyTorch neural hedging models with Emeric Battaglia, studying dynamic training batches and parallel Heston simulation under CVaR and entropic risk objectives. A matched preliminary experiment reduced peak CPU RAM by 63.8% with comparable observed test CVaR. Deep Bellman Hedging and portfolio compression are current research directions. Code and notebooks.

Gaussian-Mixture Score Stability 2026–Present

Study sharp stability bounds with Knut Sølna to quantify how errors in mixture parameters affect score estimates and Langevin sampling. Combine analytic estimates with numerical experiments.

Multilinear Convolution on the Hypercube Manuscript in preparation

Investigate sharp exponent conditions for convolution inequalities on binary cubes with Lekha Patil. Current draft: *Exact Regime for Multilinear Young Convolution on the Hypercube.*

Polynomial Approximation with Monotone Weights Manuscript in preparation

Construct counterexamples with Paata Ivanisvili for weighted polynomial approximation and examine their extensions to weighted L^p spaces.

Drug and Vaccine Safety Signal Prioritization 2025–2026

Build Python workflows for FAERS and VAERS using logistic regression, gradient boosting, calibration, and temporal evaluation. In the final VAERS test, capture 50% of eligible next-quarter statistical threshold crossings with 47.4% fewer candidate pairs for review than threshold ranking. Code, results, and presentation.

Earlier Research Projects

Mathematical Objects in Virtual Reality 2020–2022

Created JavaScript/Babylon.js visualizations with Peter Jipsen to help students explore lattices and other abstract mathematical structures in a browser and virtual reality. Project repository.

The Mathematical Basis of General Relativity 2021

Explored mathematical foundations of general relativity with Mihaela Vajiac through Chapman University's student–faculty research program.

Quantum Trajectories for Qutrit Processors 2021

Investigated quantum trajectories with Justin Dressel through Chapman University's student–faculty research program.

Road Map to Zeta ζ and Theta θ 2020

Studied zeta and theta functions with Ahmed Sebbar through Chapman University's student–faculty research program.

Brain Institute – Chapman University Summer 2018

Contributed to the IRB team for a sensory-deprivation float-pod experiment and its connection to the placebo effect. Gathered statistics on bicycle and e-bike safety and health benefits for a study of commuting alternatives.

Teaching Materials and Student Research

- **MATH 178 – Mathematical Machine Learning Lab** (Winter 2026). Prepared Jupyter notebooks and exercises covering model implementation, evaluation, and data analysis.
- **Deep Hedging Directed Reading Program** (Fall 2026). Developed accessible notes, presentations, and Python notebooks connecting financial models with numerical experiments.
- *Preserving Vector Norms Using Subgaussian Random Matrices* (Spring 2026). Mentored student contributors Nguyen Tran, Da Min Choi, Alan Li, Hanyun Wang, and Keven Zhuang.
- *Rademacher Random Matrices as Near-Isometries on the Unit Sphere*. Supervised student contributors Aparajita Bandopadhyay, Madeline Chu, and Collin Cheung.

Attended Workshops

- *Mathematical Foundations of Data Science*, Université de Montréal, May 20–24, 2025.
- *22nd Workshop on Applications and Generalizations of Complex Analysis*, University of Aveiro, March 28–29, 2025.
- *Heat Flow, Two-Point Inequalities, and Beyond*, UC Irvine, October 30–November 3, 2023.
- *SYCO 4: Fourth Symposium on Compositional Structures*, Chapman University, May 2019.

Additional Activities and Service

- Helped organize technical details for a mathematics conference, including the abstract book, participant communications, and name tags.
- Served as Math Club President and previously Vice President at Chapman University.
- Participated in the Physics Club and Philosophy Club at Chapman University.

Technical and Language Skills

- **Programming:** Python, C/C++, SQL, R, JavaScript, Java, MATLAB, and Mathematica.
- **Scientific computing and ML:** PyTorch, NumPy, pandas, Matplotlib, scikit-learn, Jupyter, Monte Carlo simulation, optimization, kernel methods, temporal evaluation, and calibration.
- **AI-assisted research:** Use LLMs as conjecture engines and coding assistants; evaluate proposed results through mathematical proofs, counterexample searches, and reproducible numerical experiments.
- **Languages:** Fluent in English, French, and Hebrew; also speaks Russian and some Arabic.
- **Additional technical skills:** Linux, Git/GitHub, 3D modeling and printing, CNC, and soldering.
- **Interests:** Weightlifting, LEGO building, and karaoke.